

Developing a Product Hypothesis and User Personas

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Welcome to Your Course

Video Transcript

It's great timing to be learning product management skills because the world has really shifted towards being design- and product-driven, both at big companies and at startups. Your job as a product manager will be threefold. The first is to articulate what a winning product looks like. The second is rally the team to build it. And then the third is to iterate until you get it right. We will cover each of these.

I'll start with the first. In its simplest form, articulating a winning product means you can craft a simple statement that says, "We are going to help Person X solve Problem Y by doing Z"; a specific customer, a specific problem, and a solution that you have defined. Now, the problem is that your natural instinct is usually to jump to the solution and build a great product. But that can cause problems.

A famous example is the NASA space pen. NASA spent a million dollars developing a pen that would work in zero gravity in space for astronauts. And the problem is, that was wonderful, but the Russians sent a pencil and it worked just as well, because they didn't sit down first and say, "What is the person and what is the problem we're trying to solve?" If you jump to the solution, you might end up with a space pen. So I'm going to help you fight that instinct. Let's get started.

Course Description

Developing and launching new products involves more than a great idea. In today's fast-paced digital economy, understanding your target customers and their specific needs is essential to launching and managing successful products. In startups and established companies alike, the product manager serves as a key player in taking a product from idea to launch to success.

To bring viable products to market, a successful product manager needs to balance creative inspiration with a disciplined approach. Articulating a winning product means you begin with a hypothesis: You need to craft a simple statement that says, "We are going to help Person X solve Problem Y by doing Z," thus defining a specific customer, a specific problem,



and a solution. In this course, you will do just that: talk to customers, define problems, and come up with a product strategy and a plan for how you would actually build that product.

In this course from Keith Cowing, Visiting Lecturer at Cornell Tech and vice president of product management at Flatiron Health, you will map your customer's journey and articulate user personas. In so doing, you will position yourself with a much clearer picture of the real problem your product is trying to solve, paving the way for a strong product strategy.

What you'll do

- Map the customer's journey
- Define and test a hypothesis with your target audience and/or customers
- Create a list of target customers, prep questions for user interviews, complete five interviews, and update the customer journey based on your findings
- Write user personas based on patterns found in the user interviews

Faculty Author



Keith Cowing is a Visiting Lecturer at Cornell Tech and vice president of product management at Flatiron Health, a technology company focused on accelerating cancer research. He has spent 15 years building products, teams, and companies, and now teaches about careers and product management. Mr. Cowing's blog can be found at [KeithonCareers.com](https://keithoncareers.com).

Keith Cowing

Visiting Lecturer

Cornell Tech

Cornell University

Prior to joining Flatiron, Mr. Cowing was a product manager at Twitter, LinkedIn, and Goldman Sachs. He was also the founder and CEO of Seamless Receipts. Mr. Cowing led the team, raised venture capital, managed the development and launch of new products, and secured business from global brands such as Oakley, Tumi, Burton, and Lenox.

Mr. Cowing received an MBA with distinction in entrepreneurship and a Bachelor of Science cum laude in electrical and computer engineering from Cornell University.



Cornell University

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Read: Preview Your Course Project

Throughout this course, you will practice applying the teachings of each module in a multi-part course project. The final project part, in which you will synthesize all course findings, is worth a significant portion of your grade. You can review all information on assignments and grades on the **Grades** page of this course.

Since each module builds on the previous ones, it is important that you work through the course in the order it appears. **You must submit all prior assignments to access the final project part.**

Note: Though your work will only be seen by eCornell, you should take care to obscure any information you feel might be of a sensitive or confidential nature.

Course Project Parts

Preview the questions you will answer in the course project in the sections below. As you work through the course, reflect on how these questions relate to the course content and to your own experience.

— Course Project, Part One — Mapping Your Customer's Journey

The purpose of this course is to help you clearly define your target customer and the problem your product will solve. You will accomplish this by mapping your customer's journey, selecting a problem on which to focus, performing user interviews, and writing user personas. The course project will guide you through these steps via real-world practice.

In this first part of the course project, you will start with a description of the problem space by mapping the customer's journey and by crafting a simple hypothesis. You will then define a specific customer, a specific problem, and a solution.

— Course Project, Part Two — Conducting User Interviews

☆ Key Points

You will apply the content from this course in a multi-part project

You must submit all prior assignments to access the final project part



In this part of the course project, you will be conducting user interviews. By talking with target customers, you will learn what truly matters to them, identify patterns, and begin to form the foundation for your product strategy. Make sure to use the same product example you used in Part One.

Note: You will need to interview five potential customers for this part of the project. Be sure to plan ahead accordingly.

— Course Project, Part Three — Articulating User Personas

In the third and final part of the course project, you will write a proposal to your manager outlining your suggestions for managing the bottleneck you identified in Part Two of your course project. Your solution must demonstrate a way to improve the bottleneck without hiring new employees, purchasing more equipment, or increasing resources in any way. You will not have to submit your proposal at your job to receive credit.

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Module Introduction: Mapping Your Customer's Journey



You want to sharpen your skills as a product manager or completely make the switch to become one. Have you thought about where to start? In its simplest form, articulating a winning product means you begin with a hypothesis. You need to craft a simple statement that says you are going to help Person X solve Problem Y by doing Z: You define a specific customer, a specific problem, and a solution.

In this module, you will hear from Mr. Cowing about the steps involved in becoming a more successful product manager. You will start by mapping the customer journey and choosing the right problem for your product to address; in other words, you will practice how to help Person X solve Problem Y by doing Z. Through such practice, you can position yourself to make better decisions about your product strategy, which is anchored in improving your customer's life.

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Watch: Mapping the Customer's Journey

When we talk about the customer's journey, what does that look like? Mr. Cowing introduces you to what's involved so that you can articulate with specifics and get your team and your customers on the same page.

Video Transcript

Now we talked about crafting the simple statement: "Help Person X solve Problem Y by doing Z." The first step is to map the customer journey, and I'll give you an example by walking through a product that I worked with, LinkedIn Recruiter. If you're not familiar, LinkedIn is a professional social network where you have a representation of you, where you've worked, and a network of contacts. So we'll walk through a very simple representation of a customer journey.

LinkedIn Recruiter is a product that helps recruiters find talent. But what matters first is, what is the problem that the recruiter is trying to solve; what is their journey? And so when we walk through this, first, if somebody needs to hire somebody, the recruiter writes a job description. Then they go out and they find candidates; they may put ads on the web, send emails, make phone calls, go to job fairs. The job candidate will learn about the job, then apply to the job. The recruiter will review resumes. They will perform interviews.

If things go well, they will extend an offer, and then hopefully the job candidate will accept the offer. And that's the journey, end-to-end, in its simplest basic form. And this gives you a starting point so that you can have a common understanding and vocabulary, both for your team, so you can talk about this customer journey and the problems that you could solve, and for your customer, so that when you talk to them you can be specific and you can get everyone on the same page.

Now, some nuances when you're in the real world. If you're working on an existing product and you're building features for that product, you or your company probably have a lot of context about this and you can jump right into the customer journey and do it in detailed form. If you're building a new product but it's for the same customer, then you probably have some risks that you need to mitigate because you might have a few blind spots. But you have a head start because you already understand that customer. If it's a completely new customer — for example, when LinkedIn moved from selling to recruiters to building products to sell to salespeople — that's a big jump, and you might have to go out and talk to customers and do user interviews before you even begin mapping the customer journey.



Watch: Choosing the Problem to Focus On

When you map the customer's journey, you may discover problems to potentially address. Should you focus on just one problem or multiple problems? In this video, you will discover how you can identify which of these problems you could choose to work on solving.

Video Transcript

Now after you've mapped the customer's journey, what you need to do is select a specific problem to work on. But there's a few things that you want to think through. First of all, it needs to be a problem that really matters to your customer, that they deeply care about. The second is you want an opportunity for step change improvement; not just a 10% improvement, but a huge improvement for them.

You also want it to be something that your team or your company is well-positioned to solve and that hopefully will give you differentiation from the competitors in the market. So you want to go through this customer journey and analyze each of the problems that you could potentially solve, and then think through, "How painful is this problem?" "How frequently does this problem occur?" "What are the current alternatives that our customer could use instead?" And what skills, experience, and resources does your company need to solve it well?

I'll give you an example back to the LinkedIn Recruiter customer journey. If we think through that, there are several problems LinkedIn could have solved. They could have solved job postings and said, "We're going to make it easier for you to write a job description and post it." They could've solved managing resumes and doing interviews. What they chose was to focus in on a very specific problem, was finding relevant talent for the job, and that was the first product that LinkedIn focused on for recruiters. They eventually built a portfolio where they solve many other problems as well, But the first problem was laser-focused on "We are going to help you find talent that is out there, that is relevant to you, that otherwise you never would have found because they didn't have an active resume posting to job sites."

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Tool: Steps for Choosing the Right Problem

When you select a specific problem to address, there are some criteria you need to think through. First of all, it needs to be an issue that really matters to your customer, one that they care about deeply.

Second, you want an opportunity for step change improvement — not just 10% improvement, but a huge improvement for your customer. You also want it to be a problem that your team or your company is well-positioned to solve and that will hopefully differentiate you from your competitors in the market.

Download this tool to use now, and save it for later use to help you choose the right problem based on these key considerations.

Important Instructions for Saving Your Work:

When you click the link to open the PDF, it may open in a new browser window. Save it to your desktop or preferred storage location. After you save it, open it with Adobe Reader and save it with a new name (such as the name of this course). This will give you a working copy that you can review, print, or add content to, without losing your work. You will also retain a clean copy of the original document to use after you complete the course.

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 [Download the Tool](#)

Use this [Steps for Choosing the Right Problem](#) tool to help you get started.



Read: Companies and the Problems They Focused On

Choosing the right problem to address is a critical step en route to developing a winning product strategy. Focus too narrowly and you may never achieve the scale required to succeed. Think too broadly and your product could get lost in the noise. This is an important decision, and one that you have to make with imperfect information. So it's important to practice and learn from others' mistakes and successes.

Let's analyze the product strategies of some companies and how they combined market research, relevant trends, team capabilities, and gut instinct to set a direction.

Below are companies or products and the problems they chose to focus on, along with a debrief of each.

Read each example and consider the accompanying analysis.

☆ Key Points

As you learn and the world changes, you may need to adjust the target customer or problem.

The same strategy (target customer + problem) may lead to very different technologies or products over time.

Your team's capabilities can impact which problems you are successful at addressing.



Facebook	Share photos with your friends	Facebook focuses on many problems now. But at the very core, if you were to choose one, it's sharing photos with your friends. This is what fueled its massive growth. When Facebook started, it targeted college students and the problem of knowing their classmates' names and relationship statuses. Then Facebook noticed that people were frequently changing profile photos, so Facebook built photo-sharing features. This highlights the fact that your decision does not have to be static; successful teams learn and adjust accordingly as they go.
Hubspot	Organize your digital marketing activities that help you get new customers, including your website, blog, search engine optimization, and social media campaigns.	Hubspot noticed a gap in the market. There were expensive, powerful tools available for large companies with huge marketing budgets. There were also simple tools available for freelancers or very small businesses. Yet there were limited options for small to mid-sized businesses who couldn't afford the expensive enterprise tools but needed more than a sign outside their store. In this case, the combination of persona (marketer at a small to mid-sized business) and problem was key. Getting the wrong persona would have led to complete failure.



Google Photos

Organize, store, and share all of your photos.

Google launched Google Plus in 2011, which was a social network aimed to compete with Facebook. This product failed to get traction; however, it had photos as a key product feature. That investment in photo capabilities ultimately led to the Google Photos launch in 2015, which is now a highly successful product.

It's important to note that Google's ultimate success was aligned with its capabilities as a company: large-scale storage, efficient product design, and machine learning. Google has never been adept at social features, and this was clear with Google Plus. Different companies are skilled at solving different types of problems.

Netflix

Watch high-quality movies and shows as easily as possible.

Netflix has gone through several generations of technology. It started with DVD rentals by mail. Then it moved to streaming over the web. Then it moved to original programming, producing the content itself.

Each of these technical shifts happened over many years and resulted in completely new tactics and operations. However, the fundamental problem it focused on was still the same: helping you watch high-quality movies and shows as easily as possible.

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Discussion: Choosing a Problem and Mapping the Customer Journey

Create a post in which you share a problem that matters to a customer in your field or line of work. Explain whether your company/organization is experienced in solving it. What are the alternatives? How painful is it?

Then, respond to your classmates' posts mapping their customer journeys by completing this statement: Help person X solve problem Y by doing Z.

Instructions:

- You are required to participate meaningfully in course discussions.
- You must post to the board before seeing other students' posts. You are encouraged to respond and engage meaningfully with your peers.
- Your responses will be reviewed according to the following criteria:
 - Comprehension: Demonstrates an understanding of course content.
 - Synthesis: Demonstrates ability to synthesize course content within the project topic.
 - Originality: Utilizes your own original thinking.
 - Development: Clearly articulates and fully addresses all required questions/topics.

To participate in this discussion:

Click **Reply** to post a comment or reply to another comment. Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review [eCornell's policy regarding plagiarism](#) (the presentation of someone else's work as your own without source credit).

Please note:

While discussion board postings will be accepted through the end of the course, we strongly encourage everyone to move through this course as a group. As such, postings that are made to this board after 5pm ET on the due date will be read and graded but may not receive a response on the board from your facilitator and/or your peers. Please let your facilitator know if you have any questions.

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Assignment: Course Project, Part One — Mapping Your Customer's Journey

Please complete this activity in the course.

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Module Wrap-up: Mapping Your Customer's Journey

In its simplest form, articulating a winning product means you begin with a hypothesis. You need to craft a simple statement that says, "We are going to help Person X solve Problem Y by doing Z": You define a specific customer, a specific problem, and a solution.

You have now had a chance to do just that by choosing the problem your product will solve. You practiced articulating what a winning product looks like and completed Part One of the project in which you mapped the customer journey.

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Module Introduction: **Conducting User Interviews**



People assume they understand their customers. But to know your customers is to build empathy for them, and interviews give you the opportunity to discover what truly matters to your customers. Mr. Cowing will share tips to conduct effective user interviews to build that empathy and find the threads that matter. You will examine the art and science of interviewing and how to structure questions. Through practice, you will prepare an interview template and interview your own target customers.

Note: In the course project for this module, you will need to identify the target audience, find five people who are willing to be interviewed, write interview questions, and conduct five 20-minute interviews. You may want to plan your time accordingly.

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Watch: Goals of a User Interview

What is involved in really getting to the problem you are trying to solve? How do you uncover the clues? You should be in a position to clearly articulate the problem yourself by talking to your customers and ignoring the solution for now.

Video Transcript

Now we're going to go out and actually talk to customers and do user interviews. And before we do that, we want to clarify, what is the goal of a user interview to make sure we get what we want?

The goal of a user interview is to very clearly be able to articulate — to yourself first — who is this person, what is the problem that they're facing, and still avoid the solution for now, but just walk through this customer journey and truly understand it. What does this whole person's life look like? What things do they love, what things do they hate? What are the problems you could potentially focus on; which problems where if you solved them it would be OK but it wouldn't really change their life, and which problems would completely change everything for your customer, and have a deep understanding of that.

And if we go back to the space pen example, if NASA had gone out and done detailed user interviews, you might have sat down and said, "OK; well, this very particular type of person, an astronaut, has complex equipment and gloves, so it's difficult for them to hold a writing utensil of any kind." And the problem that we're focused on is recording data while they're on a flight, and they need to record that data and have it never be lost. But they don't need to analyze it in real time; maybe they come back to Earth and analyze it later.

So the only point is that they're trying to solve the problem of, with a lot of equipment and gloves when it's difficult to write anything and they're in zero gravity, they want to record data and never lose it. So perhaps the best solution in that situation could have just been a recording device that they can talk to and save, make sure they don't lose it, and then analyze it when they get back.

There are a variety of ways they could have approached that problem from a solution perspective. But when you do the user interviews, the things you would have uncovered is who is this person, what frustrates them, what are the things in their life that are relevant to this problem — such as the equipment and gloves that they're wearing — and what are they actually trying to accomplish; maybe it's tracking data and bringing it back. So when you go



out and you talk to users, and you do user interviews, very clearly try to uncover with them, what are the factors in their life that affect how you're thinking about this problem, and what is the problem and how much does it matter to them.

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Watch: Building Empathy for Your Customer

You may already feel like you know your customers. As you will see, however, there is always more to learn and understand.

Video Transcript

Now we're really going to dive deep and build empathy for the customer. And a natural instinct here is to assume that you understand your customer. Don't do that. And I'll give you a few examples. So at Flatiron Health, I build software for oncologists — cancer doctors — to manage their day and treat patients. And when I think about that, I'm always thinking about, how can I help doctors give better care to patients? And that's my natural instinct of what matters to them most.

But then when I go talk to doctors in the real world, what I learn is that they already feel like they have a strong grasp of how to give patients good care. What they actually want from technology is something to save them time and help them manage the finances of their cancer practice. Turns out in healthcare it's very complicated to run the money in the business, and they see 40 patients a day and so they're extremely stressed for time. And what they want is something that can help them save time and make sure that the practice is financially viable. Those are the problems they actually want you to focus on. And that's not intuitive when you go into it, so you have to get out in the real world and talk to these customers to understand it.

And at Flatiron, we actually hire doctors who are currently practicing so they can work with us and help us understand our customer better. And even they aren't completely our customer because they're the 1% that has left the system to work on the system and make it better; the other 99% actually have much more repetitive, routine views of their day. And so even the doctors that we have on staff aren't necessarily good judgments of what our customer actually needs, and we have to keep that in mind.

Another example is when I worked at Twitter. So I was a Twitter user and I thought, "Well, I can get my head around how other people might be using this product." But I had one account, and I looked through a variety of interests. It turns out that in Japan the general behavior was people had one account per interest. So they might have an account for sports, and an account for news and politics, and an account for business-related things; completely different behavior. Without getting out and talking to them, I never would have understood that. I was also a heavy consumer of content; I would read a lot but I wouldn't post much.



And then there are some people that did a lot of talking and a lot of production of content; very different type of user. And I had to go talk to those people to be able to build effective products to help them.

And so you may be a customer, but you are always biased. You might be biased by your location of where you live in the world. You might be biased by socioeconomic factors. Even by your device; when I worked at LinkedIn, we noticed that a lot of the folks on the team had iPhones. And that's a device bias where we had to be really careful to think about folks that were using Android or other devices. And so assume you're not the customer, and then get outside the building and go talk to people, and that's when you'll get a true sense, either within your country or around the world, how people might use this product and how to really design a solution that works for them.

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Watch: The Art and Science of Asking Questions and Listening

Going through the motions of interviewing is never enough. There are some core principles to understand; by following them, you'll get the information you need to formulate a product strategy.

Video Transcript

Now when you actually go out and talk to customers, just getting in front of them and doing an interview is not enough. Not all interviews are created equal. So you actually have to build up the skills and the practice to make sure that you get the information that you need. There are some core principles that we want you to follow.

The first one is don't lead the witness. It is so easy to direct people in the direction that you want them to go, even without realizing it. And so what we did at Twitter is we had a listening lab, where we would have people use a product and somebody would walk them through it. And as a product manager, we weren't even allowed to be there and talk to them; we had to watch on live video.

And what that had was a situation where you realize when you were sitting down and showing your phone to somebody, without realizing it, you would coach them through the flow, and you would say, "Well, click right here," and "You should really do this thing," and "You should really do that." You were constantly pushing them in the direction you wanted them to go. You'd give that same product to somebody in a listening lab and you couldn't speak. And it was so frustrating because you thought it was so easy to use a product; it turns out they had no clue what to do, and then you had to revisit your design.

So you know what? Let's start over. And so don't lead the witness. It's incredibly important to ask a question and then just stop talking and let them get their last sentence out. Let them think out loud and walk through whatever is in their mind. It will take them a minute to articulate it, but let them formulate their thought and let them tell you. Don't lead the witness.

Another point is that it's an art and a science of interviews, and you have to blend both. So the art is mixing in open-ended questions and closed-ended questions. You might have an open-ended question like, "Walk me through your day." You might have a closed-ended question like, "Well, do you take a train to work?" Maybe that's a very specific and important point for the product that you're working on.



And while you're having this conversation, build empathy, understand who this person is, and find the threads that really matter to them and then go deep there. So you start broad, you find the key points, and then you go deep and you ask specific questions. And that's what will uncover these nuggets that will help you develop these products that really change their life. And there's a science to it as well; there's a repeatable, disciplined process.

In *The Sprint Book*, they talk a lot about Michael Margolis, who's one of the best user interviewers out there. And he will take very detailed notes. He'll bring mints before every user interview to make sure that there's nothing distracting like his breath. He'll mirror their body language as he's in the room with them to build a little bit of rapport, and then either have a friend take notes or have a recording device so he doesn't have to be distracted by taking notes. And so when you blend this art and the science together, that's when you can truly have a productive user interview so that you come away with the information that you'd need to formulate a product strategy.

Note: In the course project for this module, you will need to identify the target audience, find five people who are willing to be interviewed, write interview questions, and conduct five 20-minute interviews. You may want to plan your time accordingly.

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Watch or Listen to Good Interviewers

Talking to customers is one of the most informative activities in which you can engage. This applies throughout the product life cycle, whether you are starting with just a product hypothesis or updating a mature product that has been in the

market for years. But simply talking to customers is insufficient, since not every interview is created equal. You need to build the skills to facilitate a conversation that produces valuable and actionable information. Interviewing is a craft, and to improve your skills it is helpful to study how professionals conduct interviews.

For this activity, you will spend time watching or listening to interviews and observing both what the interviewers do well and where they could improve.

Here are a few things to consider:

- **How do they build empathy with the interviewee?**
- **What kind of information did they research prior to the interview?**
- **When do they use closed-ended questions (e.g., “Do you take the train to work?”) vs. open-ended questions (e.g., “Can you tell me about your commute?”)?**
- **When do they set context before asking a question and when are they direct and succinct?**
- **What percentage of the time does the interviewer speak vs. the interviewee?**
- **Does the interviewer jump in to fill silences or patiently let the interviewee finish their thoughts?**

Over time you will develop your own interview style. As you do, continue observing how other people conduct interviews so you can incrementally pick up tactics to improve your interactions. Below is a sample list of interviewers to get you started, but feel free to expand beyond this list. The most important thing is to find three to five people that are effective at interviewing and covering interesting subjects, watch or listen to them, and take notes.

Though this activity does not require formal submission, it will inform the work you do in the course project.

Instructions:

 [Download the Tool](#)

Use the [Watch or Listen to Good Interviewers](#) template to help you note key points.



Download the [Watch or Listen to Good Interviewers](#) template to record your observations.

1. Watch or listen to videos of good interviewers; a few suggestions are provided here. You may also find topics you are interested in and observe what makes them effective. *Note: If the link is not working, open your preferred podcast or video app and search for the interviewer.*

- [Guy Raz, NPR's How I Built This](#)
- [Jason Calacanis, This Week in Startups](#)
- [Emily Chang, Bloomberg Studio 1.0](#)
- [Peter Kafka, Recode Media](#)

2. Take notes on what makes these interviewers effective as well as what they could do to improve.

Important Instructions for Saving Your Work:

When you click the link to open the PDF, it may open in a new browser window. Save it to your desktop or preferred storage location. After you save it, open it with Adobe Reader and save it with a new name (such as the name of this course). This will give you a working copy that you can review, print, or add content to, without losing your work. You will also retain a clean copy of the original document to use after you complete the course.

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Read: Writing Effective Interview Questions

When interviewing target customers, how you ask questions will impact your speed of learning. Consider each question below and ask yourself, "How could I ask this question in a better way?"

Try to avoid "leading the witness" by phrasing your question in a way that gives the other person hints about what you expect them to say. And when you ask a question, dig deeper by following up with a "Why?" This is an effective strategy for yielding meaningful results and insights that you might not have elicited otherwise.

Learning to prepare effective interview questions will help you generate productive conversations with customers, which will ultimately lead to better product strategies.

Quick Check

For each of the sample questions below, think of a stronger way to rephrase. When you think you have an answer, review Mr. Cowing's suggestions.

▼ "What's the most important attribute in a new computer; the price?"

Stronger rephrasing: "Which attribute of a new computer is most important to you? Why?"

Instead of "leading the witness" by mentioning price, you can ask an open-ended question in which the other person comes up with their own answer. And by following up with a "Why?" you may learn important additional details that you hadn't considered.

▼ "What kind of laptop do you use?"

Stronger rephrasing: "What devices do you generally use on a daily basis?"

☆ Key Points

Leverage open-ended questions, especially early in a topic discussion.

Do your homework so you can skip basic questions and get to details faster.



Instead of just learning what brand laptop they use, you might find out that they frequently swap out a tablet for a laptop, or have a desktop at work so the laptop is only for traveling, or they are using a personal digital assistant. Much of the information will be irrelevant, but some hidden pieces of information will change the way you think about your product and your customer. So make sure you don't constrain the conversation accidentally. You can always follow up with a quick "What kind of laptop do you use?" if you didn't get a specific answer that you need.

▼ "Do you find it challenging to find jeans that fit well, like many other people do?"

Stronger rephrasing: "How do you shop for clothes?"

Stronger rephrasing: "How do you find clothes that fit well?"

Stronger rephrasing: "Do you find it challenging to find jeans that fit well?"

If you have a hypothesis (i.e., "It's challenging to find jeans that fit well"), then it is easy to lead your interviewee to nod and confirm. Remember, they have nothing at stake here, so the natural tendency is to tell you what you want to hear. Start broader at first so you get the full context and story. Then you can dive into your question and specific hypothesis, but phrase it in a neutral way that doesn't lead them to one side or another.

▼ "Where did you grow up?"

Stronger rephrasing: "I noticed on your bio that you grew up in a small town outside of London. Can you tell me what that was like?"

You will not always have the opportunity to do this, but the higher-stakes the interview, the more you want to do your homework in advance. Collecting information ahead of time allows you to spend the same amount of time with the interviewee but get much richer answers.

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Read: Example Interview Template

Now that you have a better understanding of writing effective questions and conducting strong interviews, how do you organize your questions to set yourself up for productive conversations?

It helps to structure your interview with a simple template that you can use with each customer. The script will help you remember all of your questions and organize the conversation into themes. With practice, you will get a sense of what themes are critical to your product area. Having a script will assist you in managing this flow through the conversation, but you don't need to stick to it exactly. Just make sure to update the script as you learn from your conversations.

Be sure to take detailed notes. You can make these observations yourself, have a friend help, or record the conversation with the interviewee's permission. These notes can be invaluable down the road! Also, when you hit on something important, like their satisfaction with their current alternatives, follow up and ask why. Don't be afraid to ask "Why?" multiple times until you understand the root cause of the issues they're facing.

Example

Let's say you are considering building a new service that helps chefs order food for their restaurants. You have an opportunity to interview three to five chefs in a local city. The following is an example of a potential interview script. After a couple of interviews, you will likely discover things that are important and not listed on your initial draft, at which point you simply update the script for future interviews.

Introduction

Thank you so much for talking to me. We are doing research to investigate how we might provide a better solution to help you order food for your restaurant. I will be taking notes as we talk and will be as efficient as possible. Should we get started?

Topic 1: Types of food

☆ Key Points

Structure your interview using a template.

Organize themes critical to your product area.

Take detailed notes.

Questions:



What range of food do you offer at your restaurant (e.g., vegetarian, fish, red meat, pork, poultry)?

How many different food suppliers do you use? Which suppliers are they?

Notes:

Topic 2: Ordering behavior

Questions:



Who places the food orders?

How frequently do you order?

What is your expected time from order to delivery?

How do you manage inventory?

Notes:

Topic 3: Current alternatives

Questions:



What do you use today for your food ordering?

How satisfied are you with your current setup?

What causes you the most pain in the ordering process?

If you had a new solution for ordering food, what would be most important to you?

Notes:

Topic 4: Technology

Questions:



What do you use for entering the orders (e.g., phone, laptop, tablet)?

What could make this more convenient for you?

Notes:

Closing

Thank you so much for talking to me. We are doing research now and plan to build a prototype over the next two months. I don't need to take any more of your time, but since you talked to us, I will be happy to update you when the product is ready and, if you'd like, give you early access to use it and provide feedback.

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Tool: Interview Template

Organizing your thoughts is the first step in laying out the direction of a productive interview. Use this template to help you develop a script before the interview and record notes during it.

 [Download the Tool](#)

Use this [Interview Template](#) to help you achieve an effective interaction.

Important Instructions for Saving Your Work:

When you click the link to open the PDF, it may open in a new browser window. Save it to your desktop or preferred storage location. After you save it, open it with Adobe Reader and save it with a new name (such as the name of this course). This will give you a working copy that you can review, print, or add content to, without losing your work. You will also retain a clean copy of the original document to use after you complete the course.

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Assignment: Course Project, Part Two — Conducting User Interviews

Please complete this activity in the course.

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Discussion: Sharing Your Interview Results

Create a post in which you share the results of your customer interviews. For which scenario were you interviewing? Explain your experience and what you learned from conducting user interviews. What challenges did you have? Did you find there were certain types of questions that elicited more feedback than others?

Provide suggestions for one another about ways to overcome challenges.

Instructions:

- You are required to participate meaningfully in course discussions.
- You must post to the board before seeing other students' posts. You are encouraged to respond and engage meaningfully with your peers.
- Your responses will be reviewed according to the following criteria:
 - Comprehension: Demonstrates an understanding of course content.
 - Synthesis: Demonstrates ability to synthesize course content within the project topic.
 - Originality: Utilizes your own original thinking.
 - Development: Clearly articulates and fully addresses all required questions/topics.

To participate in this discussion:

Click **Reply** to post a comment or reply to another comment. Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review [eCornell's policy regarding plagiarism](#) (the presentation of someone else's work as your own without source credit).

Please note:

While discussion board postings will be accepted through the end of the course, we strongly encourage everyone to move through this course as a group. As such, postings that are made to this board after 5pm ET on the due date will be read and graded but may not receive a response on the board from your facilitator and/or your peers. Please let your facilitator know if you have any questions.

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Module Wrap-up: **Conducting User Interviews**

As you have discovered, in order to identify what matters to your customers, you have to talk to them and conduct effective interviews with them. In this module, you explored the importance of writing good interview questions. You examined the art and science of asking questions and listening. You wrote example interview questions and completed Part Two of your course project, in which you practiced conducting user interviews.

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Module Introduction: **Articulating User Personas**



You will now have an opportunity to articulate user personas and identify patterns in your customers' journeys. You will have an opportunity to create well-written personas, which will help you articulate how people will value and use your product. Keith Cowing will guide you through key considerations so you can better define user personas that will lead to effective product decisions. Through practice, you will begin to think of the world in terms of personas.

User personas are powerful tools; they impact your fundamental strategy and value proposition, your product roadmap and features, and even how you will talk about the product. You'll now have an opportunity to discover how to articulate user personas and identify some of the patterns in your customers' journeys. You will receive a checklist, which will help you articulate how people will value and use your product. You will examine some of the key considerations that help yield the kind of user personas that lead product managers to make more effective product decisions.

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Watch: Personas Are Powerful

In this video, Mr. Cowing introduces you to user personas and how they can help you focus your energy into the right things.

Video Transcript

Now let's talk about user personas. A persona is a description of a type of customer based on their characteristics — could be demographics or their behaviors — their motivations, and their goals. And a well-written persona will help you understand how somebody will value your product and how they will use it.

So back to the LinkedIn Recruiter; let's dive in a little deeper there. LinkedIn Recruiter is actually a marketplace so it has recruiters and candidates, which means we have two sets of personas. On the recruiter side, there's a few different types of recruiters. There's different ways to slice and dice this, but one way is to say, well, there's retained search, which is the really high-end recruiters that would hire senior executives like CEOs and SVPs. It's high touch, it's confidential search, and they're paid a large upfront fee whether they actually close the person or not.

Then there are contingency search recruiters, and they hire engineers and salespeople; folks like that. It's a medium-touch process, and they're paid a commission so maybe they get 20% if they close a hire. And then there are in-house recruiters. These are people that work for the company — not an agency that's being hired — and they go out and they try to close candidates, and they may get paid a salary and perhaps a bonus based on the number of people they hire.

Then on the talent side, the candidates, let's say there's folks that are unhirable; they love their job so much they will never move, or there's some particular situation — they care about their commute more than anything; their job's across the street so you're just not going to be able to hire them, no matter what. So there's the unhirables. Then there's the active candidates; these are people who are either employed or unemployed, but they're actively looking for a job. They have a resume that's filled out, they're going to job fairs, they're submitting their resume, they're applying to jobs. And then the last one is passive candidates.

So we talked a little bit about where they are happy in their job, they don't have a resume, they're not putting it out on the street, but if you call them with something that really could



give them a great opportunity in their career, they will hear about it and they could be interested.

And so when we dig into this, LinkedIn Recruiter ended up being a wonderful solution to start with for in-house recruiters and passive candidates. So an in-house recruiter can find passive candidates they never would have had access to before, and they can save the 20% commission they would have spent on contingency search otherwise. Now we're getting much more specific about what the product needs to do. And this impacts your fundamental strategy.

So when you think about the roadmap and the features, now we need passive candidates to have a full profile in order for people to be able to search and find them. So then you find features on LinkedIn like showing what percent complete your profile is, and it's 60% complete and 80% complete, and that's a factor to get people to try to fill out more fields so that recruiters can find them.

And also job notifications when somebody changes jobs. If you get a notification that somebody is in a new position, that's interesting and that may help you make sure that you update your job if you ever change positions, which means your LinkedIn profile's up to date, even if your resume is not, so that these recruiters can find you. And when you go out and you talk to big companies selling LinkedIn Recruiter, the old story might have just been, "Hey, we're building a product to help you hire talent." And that's OK, but a lot of people pitch that.

Now you can make it very specific, and you can go out and say, "20% of the people that you want to find are unhirable; forget about them. 20% are active candidates; you already have them in your pool, so you already have them. But 60% are passive, and today you're missing most of them. Now for the first time ever you can search all passive candidates, save on the commission, and if you find one hire, this product will pay for itself." Now they get it; now they're excited. And so these personas enable you to focus the product, focus your energy, and align your whole organization.

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Watch: Writing a User Persona

Now we take user personas a step further: Here you'll discover what to do with the notes from your customer interviews.

Video Transcript

Now let's write a user persona. Back to the definition; a persona is a description of a type of customer based on their characteristics — such as their demographics or their behaviors — their motivations, and their goals. And if it's done well, it's written in a way that helps you understand, how are they going to get value out of your product and how are they going to engage with it.

So the first step is you want to start organizing your notes from your interviews, and start to identify what are the characteristics, what are the motivations, what are the goals of these people, and can you identify patterns. So, for example, you may identify demographic patterns. This is easy, it tends to be crude, but sometimes mass marketers will use this. Zip code, for example, is a way to differentiate types of people around a country or around the world. But the reality, especially software products, tends to be much more nuanced and specific. So think deeply about their lifestyle; what do they love, what do they hate, what are their very specific goals and motivations.

And then you can determine the factors that matter for your product and the factors that are irrelevant. And you can begin to divide your users into personas and name them so you can have specific conversations with your team. For example, if you're selling home security video equipment, you may have an urbanite persona, who is a person that lives in a high-rise in a big city. And you may have a suburban family persona, which is a family that lives in a rural environment. And the products that you sell to them may be very different. Or it may be the same product but a different presentation and set of things that you talk about in order to get them to buy it.

At LinkedIn, we talked about active candidates versus passive candidates, which wasn't about their goals — I mean, it wasn't about their characteristics or where they live; it's about their goals and motivations. An interesting business case is pregnancy tests. So when you go to the store, you will see two different boxes sometimes; the same exact product underneath. But one will be in a box that's bright blue and pink and has pictures of babies, and this is for the hopeful mom who wants to be pregnant. And right next to it is a very subtle box; it's



black with white text and nobody knows what it is when you purchase it, and that's for the nail-biter persona who doesn't want to be pregnant, has different goals and motivations.

So the things that you're looking for in personas are very product-dependent. And after you define the personas, you want to make sure that they're realistic so that you can identify these people in the real world. Because if you can't identify who is with what persona, then it's not useful for you and your product. For example, at Twitter, we had creators and we had consumers; people that created a lot of content and tweeted a lot, and people that consumed a lot of content and read a lot. And the way that we differentiated those people was just by looking at the data and the usage, and you could clearly differentiate who falls into what category, and then you could build features that applied to each of them.

If you think about a hotel, maybe they have a persona that's a luxury business traveler and another persona that's a budget-conscious family on vacation. Well, how do you know the difference between those two? Maybe you can tell when they walk in the door, or maybe you know when they book. Because people who book through an American Express corporate site tend to be a business traveler, and people who book through Expedia and have four people coming tend to be the budget-conscious family. And so think about what's realistic within your personas of how to identify them in the real world, and this is where you blend that art and that science to really differentiate the types of people that you're targeting and how to build great products for them.

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Tool: Guide For Defining User Personas

As you define the user personas of your customers, it is important to be realistic so that you can identify these people in the real world. Personas identify the characteristics, motivations, and goals that groups of customers have in common and impact how they interact with your product. Use this tool to help you find the patterns that will matter for your product, enabling you to write effective personas for your business.

 [Download the Tool](#)

Use this [Guide For Defining User Personas](#) to help you craft effective personas.

Important Instructions for Saving Your Work:

When you click the link to open the PDF, it may open in a new browser window. Save it to your desktop or preferred storage location. After you save it, open it with Adobe Reader and save it with a new name (such as the name of this course). This will give you a working copy that you can review, print, or add content to, without losing your work. You will also retain a clean copy of the original document to use after you complete the course.

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Watch: Diving Into the Mindset of Your Personas

What's the difference between being customer-driven versus product-driven? As Mr. Cowing describes, understanding the difference will help to build great product experiences.

Video Transcript

Once you have your personas defined, you want to dive into the mindset of your personas and really build empathy for these people, who they are, and feel their pain. You want to be customer first, not product first; that's how you delight your customers and build a great experience.

So I'll give you an example. It's actually outside of the technology world; it's my travel bag, my overnight bag, made by a company called Ogio. And so if you look through the features that they have, the first one is the wheels. Instead of using small, black, nice-looking wheels, they use big rollerblade wheels, which means I can fly off curbs and run through airports. I've been through a tremendous amount of travel, and those wheels can turn corners like no other product I've ever used.

It has handles on every side of the bag, so that whether you're pulling it out of an overhead compartment in a plane, the back seat of a taxi, somewhere awkward in a hotel room — no matter what you can have a backpack in one hand and easily grab that with your other hand; with one hand, any situation, grab it and go. It has pockets that are very carefully designed for things like security screening at an airport, where I can stand up, empty my pockets, and dump my change or my wallet — anything I need to get rid of — into the top pocket on the bag without even bending over.

The material was very carefully chosen where it's somewhat flexible but it has structure. Which means that it has structure which makes it easy to pack, but when I stick it in the overhead compartment, if it's a little bit small, it will actually squish and fit in. And the design of it generally is professional but unique enough that I can always tell which one's mine when it comes off a baggage carousel.

And so when you look at this, none of these features were designed because a designer put the product up on a pedestal and said, "How do I make this bag better?" That was not their process. What they did was they followed travelers around and said, "What pain do these people face? What are all the problems that they have? And how do I solve those problems one by one?" And that's how they ended up with a delightful experience.



And the final point there is to actually think about what people call product/product manager fit. And so if somebody is a product manager on this Ogio bag and they love travel and they love travelers, they don't have to be an extreme traveler, but they have to really want to build empathy for this person. Because if they go out and follow and that's fun for them, watching travelers and seeing what they experience and trying to make that experience better, then they'll build a great product. If they hate travel, then, yes, you can make it work; you can self-motivate. But it's not going to be as exciting for you and you won't build as delightful of an experience.

So early in your career you probably want to focus more on product skills, and then later in your career you really want to focus on working on products where you are legitimately enthusiastic about the problem and about the persona of who is using this product. Because when you're customer-driven and not product-driven, you will build great product experiences.

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Tool: Action Plan

If you find it helpful to do so, you can use this action plan to guide your efforts within your own organization. The action plan on this page follows traditional SMART methodology to help you identify steps to take on the job that are specific,

measurable, action-oriented, realistic, and time-based. You may choose to use it now as a tool for yourself, as a means of demonstrating to your manager or to peers how this course will influence your efforts on the job, or you may choose to save it and use it to guide your future work.

Important Instructions for Saving Your Work:

When you click the link to open the PDF, it may open in a new browser window. Save it to your desktop or preferred storage location. After you save it, open it with Adobe Reader and save it with a new name (such as the name of this course). This will give you a working copy that you can review, print, or add content to, without losing your work. You will also retain a clean copy of the original document to use after you complete the course.

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 [Download the Tool](#)

This **action plan** can guide your efforts in developing a product hypothesis and user personas.



Module Wrap-up: **Articulating User Personas**

As you have seen in this module, user personas are powerful tools that can influence your fundamental strategy and value proposition, your product roadmap and features, and even how you talk about the product. In this module, you were introduced to the instructions necessary for writing user personas. This module also offered a tool to help you in that quest.

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Assignment: Course Project, Part Three – Articulating User Personas

Please complete this activity in the course.

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Read: Thank You and Farewell

Congratulations on completing "Developing a Product Hypothesis and User Personas."

I hope that you now feel that you are more aware of the skills that are critical to map your customer's journey, conduct customer interviews, and articulate user personas. You should now be equipped to better understand your own strengths as well as the areas you should develop to become an effective product manager.

I hope the material covered here has met your expectations and prepared you to achieve your own product management goals.

From all of us at Cornell University and eCornell, thank you for participating in this course.

Sincerely,

Keith Cowing



Keith Cowing
Visiting Lecturer
Cornell Tech

